EXPT 1 :Simulation-Based Electromagnetic Analysis of TEM, TE and TM Propagation Modes in Microwave Transmission Structures

AIM: To simulate and analyze a 50 Ω microwave transmission line and two-port network using MATLAB by determining the S-parameters (S11, S12, S21, S22), return loss, insertion loss, transmission efficiency, and optimum operating frequency over the frequency range of 8–12 GHz

Editor — untitled1MW.m

/MATLAB Drive/untitled1MW.m

```
clc;
clear;
close all;
c = 3e8;                % Speed of light (m/s)

f = 10e9;               % Operating frequency = 10 GHz

a = 22.86e-3;           % Waveguide width = 22.86 mm
b = 10.16e-3;           % Waveguide height = 10.16 mm

L = 100e-3;             % Waveguide length = 100 mm

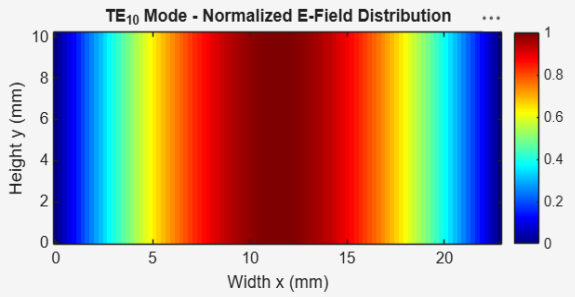
fprintf('================================================\n
fprintf(' MICROWAVE ENGINEERING - EXPERIMENT 1\n');
fprintf(' TE10 AND TM11 MODE ANALYSIS\n');
fprintf('================================================\n

fprintf('Given Parameters:\n');
fprintf('Operating Frequency = %.2f GHz\n',f/1e9);
fprintf('Waveguide Width     = %.2f mm\n',a*1000);
fprintf('Waveguide Height    = %.2f mm\n',b*1000);
fprintf('Waveguide Length    = %.2f mm\n\n',L*1000);

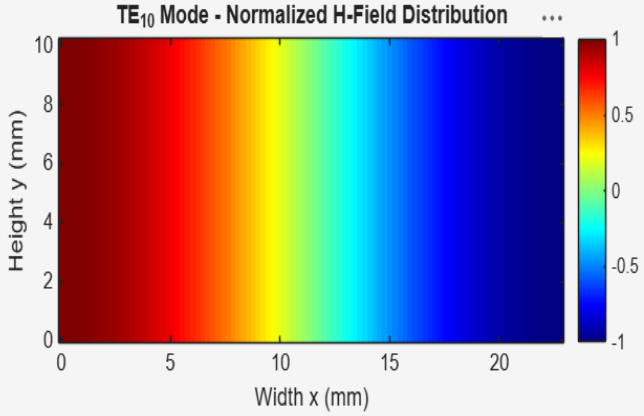


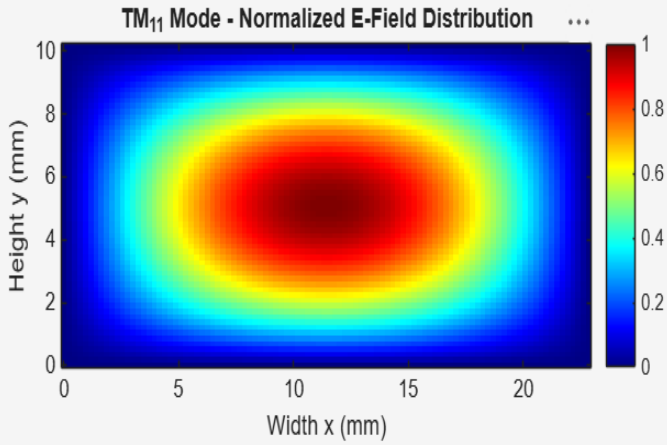
m1 = 1;
n1 = 0;
```

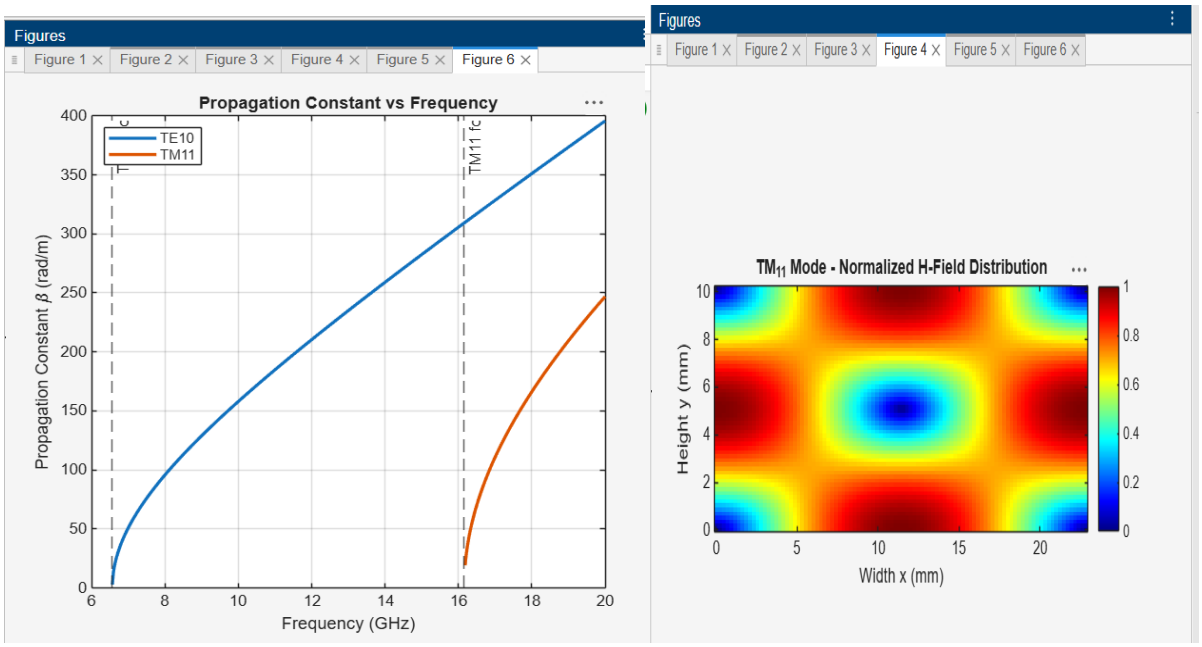
Figures — Figure 1



Figures — Figure 2







RESULT

The microwave transmission line and two-port network were numerically simulated using MATLAB. The S-parameters were analyzed, and the return loss, insertion loss, and transmission efficiency were calculated. The optimum operating frequency was identified based on minimum input reflection and maximum forward transmission.